Electrical studies — issue package

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Calculation snapshot

Snapshot ID	6BD6E80E
Calculated	15 October 2026 at 04:00
Calculation engine	LV Design Studio 1.38.0 (built-in)
System	415 V, 3-phase + N, 50 Hz

Every document in this package was produced from this one calculation run. The snapshot ID is printed at the foot of every page; a page with a different ID is not part of this issue.

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Unresolved checks

0 fails · 35 to check or not verified — listed as the reports state them. These are open items of this issue, not hidden by it.

Study	Item	Check	Result
Protection discrimination study	INC-GF (500 A)	Downstream: GF-LTG (80 A) · In ratio: 6.25 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-GF (500 A)	Downstream: GF-SKT (40 A) · In ratio: 12.5 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-GF (500 A)	Downstream: GF-HVAC (250 A) · In ratio: 2 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-GF (500 A)	Downstream: INC-DBGF1 (100 A) · In ratio: 5 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-FF (200 A)	Downstream: FF-OFF (63 A) · In ratio: 3.17 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-FF (200 A)	Downstream: FF-IT (40 A) · In ratio: 5 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-FF (200 A)	Downstream: FF-LAB (80 A) · In ratio: 2.5 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial	**WARNING**

Study	Item	Check	Result
Protection discrimination study	INC-FF (200 A)	Downstream: FF-LTG (25 A) · In ratio: 8 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-MCC (315 A)	Downstream: MCC-WP (160 A) · In ratio: 1.97 · Fault at downstream (kA): 19 · Selectivity limit (kA): 2.5 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-MCC (315 A)	Downstream: MCC-FP (125 A) · In ratio: 2.52 · Fault at downstream (kA): 19 · Selectivity limit (kA): 2.5 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y1 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R1 (10 A) · In ratio: 10 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y2 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B6 (10 A) · In ratio: 10 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y5 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B7 (10 A) · In ratio: 10 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R3 (40 A) · In ratio: 2.5 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B3 (20 A) · In ratio: 5 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R6 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B4 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y4 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B5 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R5 (10 A) · In ratio: 10 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y8 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y3 (20 A) · In ratio: 5 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**

Study	Item	Check	Result
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R4 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B2 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y7 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-B1 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-R7 (6 A) · In ratio: 16.67 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Protection discrimination study	INC-DBGF1 (100 A)	Downstream: DB-GF1-Y6 (16 A) · In ratio: 6.25 · Fault at downstream (kA): 10.7 · Selectivity limit (kA): 0.8 · Short circuit: Partial	**WARNING**
Phase balance	—	Single-phase circuits without a phase	**1 (FF-LTG) — spread evenly, unbalance may be understated**
Phase balance	MDB-1	No phase set	**1**
Phase balance	SMDB-FF	No phase set	**1**
Power factor correction	MDB-1	Demand (kW): 451 · Reactive (kvar): 243 · Existing (kvar): — · Banks below (kvar): — · PF now: 0.88 · Required (kvar): 94.2 · Bank: 100 kvar (4 × 25) · Detuning · capacitor V: None · 440 V · Breaker · cable: 200 A · 4C × 70 mm² · PF after: 0.954 · Current (A): 712 → 658	**WARNING**

Assumptions and calculation basis

Load assessment

- Demand = connected load × demand factor for each load, summed up the network (kW and kvar separately); maximum demand kVA = √(kW² + kvar²).
- Required capacity = maximum demand × (1 + growth) ÷ design loading, from the transformer sizing (no further diversity applied).

Transformer and standby generator sizing

- Transformer per main board: maximum demand (after the planned power factor correction) × (1 + 20 % growth) ÷ 80 % design loading, next DEWA standard size (500 / 1000 / 1500 kVA).
- For that size: full-load current and main breaker, LV fault level (typical IEC 60076-5 impedance, infinite MV source), voltage regulation at the design demand.
- Bus couplers: with one transformer out, the other carries both boards up to 100 % of its rating. Duty / standby: two transformers, each for the whole load.
- Standby generator from the boards on it (share of each), plus circuits marked essential. The set must carry the running kVA and the running kW at 80 % loading: rating ≥ max(kVA ÷ 0.8, kW ÷ (0.8 × 0.8)), with rated kW = 0.8 × rated kVA unless the Generators list gives the set's kW; or larger if the largest motor, started last, would dip the voltage over 15 % (X′d 25 %).

LV distribution

- Board demand from the load flow; loading = demand current ÷ the board's rated current (when entered).

Cable and breaker sizing

- Cu/XLPE/SWA multicore cable, rated for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set); derated for air ambient (methods A–E) or ground temperature and soil thermal resistivity (D1 / D2, reference 20 °C and 2.5 K·m/W), and for grouping (the cable tray route for method E, else the circuit's own parallel runs: bunched for A–C, in ducts / in the ground for D1 / D2).
- Overload protection per IEC 60364-4-43: Ib ≤ In ≤ Iz. Voltage drop from the main board to the load within the limit. Breaker Icu ≥ the fault at its terminals.

Load flow and voltage drop study

- Balanced 3-phase load flow from the design loads: demand = connected load × demand factor, summed up the network (kW and kVAr separately).
- Voltage drop per cable: √3 · I · L · (R cosφ + X sinφ) for 3-phase, 2 · I · L · (…) for single-phase; busbar voltage = nominal minus the drops of the incomers above it. Limit 4 % from the main board to the load.
- Transformer regulation is not included (busbar voltages are measured from the main LV busbar).
- Motors: drop while starting = upstream drop + cable drop × the starter's starting-current multiple; limit 10 %.

Short circuit study

- Maximum prospective symmetrical 3-phase fault current Ik″, simplified method based on IEC 60909-0 (preliminary — not a full IEC 60909 study): voltage factor c max = 1.05, cable R at 20 °C, upstream MV network infinite, transformer correction factor KT not applied.
- Source impedance from the transformer rating and %Z (split by its X/R ratio); cable R at 20 °C and X per km; parallel runs divide the impedance.
- Breaking capacity check: the fault at the breaker terminals (supply busbar) must not exceed the breaker Icu. Fault at the cable end: 3-phase for 3/4-core circuits, line-to-neutral for 2-core.
- Motor contribution and generation are neglected.

Protection discrimination study

- Each breaker against the breaker above it. Overload (time) selectivity: In upstream ÷ In downstream ≥ 1.6.
- Short-circuit selectivity: total when the maximum fault at the downstream breaker stays below the upstream magnetic no-trip threshold; otherwise partial up to that current. Confirm with the manufacturer's selectivity tables.

Earth fault loop impedance and disconnection

- TN-S system. Earth fault loop impedance Zs = Ze (at the supply board) + (R phase + R cpc) of the circuit, at operating temperature; minimum fault current with c = 0.95.
- Protective conductor (adiabatic): S ≥ I·√t ÷ 143, t = 0.1 s when tripping instantaneously (not the device let-through energy), with I the current through the conductor checked. Parallel runs: each CPC carries an equal share of the end-of-circuit fault (identical runs bonded at both ends); a result that passes only on that assumption is marked "assumed sharing" (warning) because a fault within one run or unequal runs are not covered. The total fault current sets the disconnection check.
- Disconnection per IEC 60364-4-41 Table 41.1 (TN, U0 in the 230 V band): 0.4 s for final circuits up to 63 A with socket-outlets and up to 32 A supplying fixed equipment only; 5 s for distribution circuits and other final circuits. Circuits whose purpose is not set are treated as socket-outlet circuits. Instantaneous tripping when If ≥ Ia; below that, a 5 s circuit needs the breaker's time-current curve checked. Protective conductor checked by the adiabatic equation (k = 143).

Phase balance

- Demand kW and kvar per phase (each load's demand factor and power factor), downstream boards included: single-phase circuits on their labelled phase R / Y / B; 3-phase loads spread evenly.
- Phase current I = S ÷ U0 at nominal voltage (U0 = 240 V). Neutral current = phasor sum of the three phase currents, phases 120° apart, at fundamental frequency — triplen harmonics (e.g. from LED drivers and IT loads) add to the neutral and are not included.
- Current unbalance = largest deviation from the average phase current ÷ the average. Flagged above 10 % (app default, not a standard's limit).
- Limits of this check: demand-based at nominal voltage — not an unbalanced load flow; no voltage unbalance, no neutral voltage, no phase-specific voltage drop. Single-phase circuits with no phase set are spread evenly, which hides their unbalance; they are listed.

Busbar trunking risers

- Demand: the tap-offs' loads (a floor's board demand, or the load entered) × diversity (IEC 61439-6: 0.9 for 2–3 tap-offs, 0.8 for 4–5, 0.7 for 6–9, 0.6 for 10 or more, unless set).
- Rating: Ib ≤ In (feeding breaker) ≤ busway rating × ambient derating (≈ 1 % per °C above 40 °C).
- Voltage drop floor by floor: ΔU = √3 · I · L · (R cos φ + X sin φ), each riser section carrying the diversified load of the tap-offs above it.
- Short-circuit: the busway's Icw (1 s) ≥ the fault level at the board feeding it. Busway data: typical sandwich busway — confirm with the manufacturer.

Power factor correction

- Strategy: Central — at the main board (MDB).
- Qc = P × (tan φ1 − tan φ2) to PF 0.95, from the maximum demand (demand factors applied). Capacitors already on the drawing are included.
- Banks are sized from the bottom up: each bank only covers what the banks below it do not, so nothing is corrected twice.
- Board banks: automatic (APFC) in 25 kvar steps, rounded up. Individual capacitors: fixed, rounded down (never over-correct a motor).
- Detuned (7 %, 189 Hz) where non-linear load ≥ 25 % of demand. Capacitor rating ≥ 1.05 × U ÷ (1 − p).
- Breaker ≥ 1.43 × bank current (IEC 60831 1.3 × overcurrent and capacitance tolerance). Light-load check at 30 % of demand.

DB and cable schedules

- Schedules of the boards in scope, from the design data and the latest calculation run.

Factors and limits are the app's stated defaults unless entered for the project; confirm them against the authority's requirements and the manufacturers' data.